

EBN 111/122
ELECTRICITY AND ELECTRONICS

Chapter 3
Methods of Analysis

1

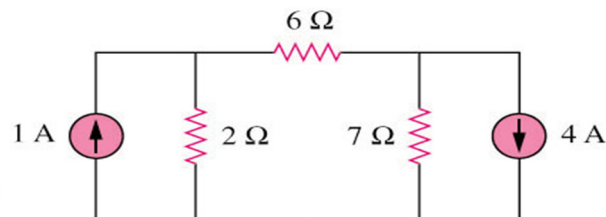
Methods of Analysis - Chapter 3

- 3.1 Motivation
- 3.2 Nodal analysis.
- 3.3 Nodal analysis with voltage sources.
- 3.4 Mesh analysis.
- 3.5 Mesh analysis with current sources.
- 3.6 Nodal and mesh analysis by inspection.
- 3.7 Nodal versus mesh analysis.

2

3.1 Motivation (1)

If you are given the following circuit, how can we determine (1) the voltage across each resistor, (2) current through each resistor. (3) power generated by each current source, etc.



What are the things which we need to know in order to determine the answers?

3

3.1 Motivation (2)

The LAWS

Things we need to know in solving any resistive circuit with current and voltage sources only:

- Kirchhoff's Current Laws (KCL)
- Kirchhoff's Voltage Laws (KVL)
- Ohm's Law

How should we apply these laws to determine the answers?

4

3.1 Motivation (3) TWO TECHNIQUES

KCL → Nodal Analysis
aka (Voltage Node Analysis)

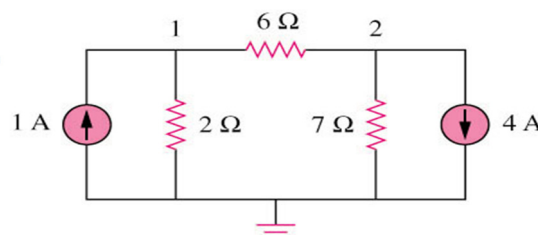
KVL → Mesh (Current) Analysis

- Can be used in any linear circuit
- To obtain a set of simultaneous equations
- Equations solved using:
 - Substitution
 - Cramer's rule (Appendix A)

5

3.2 Nodal Analysis Why?

- It provides a general procedure for analyzing circuits using node voltages as the circuit variables.
- Reduces the number of equations required to solve a circuit.



6

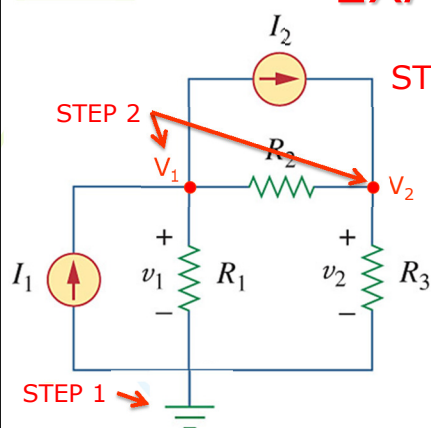
3.2 Nodal Analysis STEPS

Steps to determine the **node** voltages:

1. Select a node as the reference node (*ground*).
2. Assign voltages $V_1, V_2, \dots, V_{n-1}$ to the remaining "critical" nodes.
 - "Critical node" = node with more than 2 branches attached.
 - The voltages are referenced with respect to the reference node. I.e. Measured between Node and Ground.
3. Apply KCL to each of the non-reference nodes. Use Ohm's law to express the branch currents in terms of node voltages.
4. Solve the resulting simultaneous equations to obtain the unknown node voltages.

7

3.2 Nodal Analysis EXAMPLE 1



STEP 1: Select a Reference Node
WHERE??

- Negative terminal of V-source
- Entry terminal of I-source
- Node with many branches
- Bottom node (many cases)

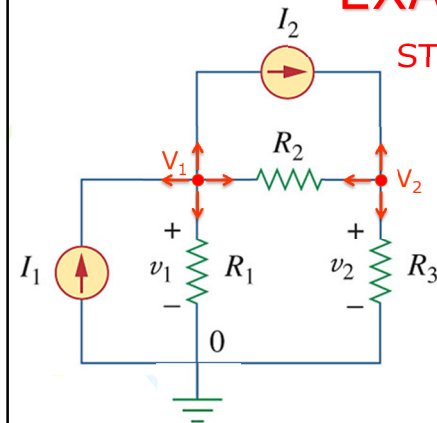


STEP 2: Name critical nodes:
 $V_1, V_2, \dots, V_n$

8

3.2 Nodal Analysis

EXAMPLE 1



STEP 3: Apply KCL to each node

- Assign a current to each branch at each critical node
- All currents to point **OUT** of the node.
- Remember: Current flows from **HIGH to LOW** voltage
- All branch currents to be written in terms of Ohm's law:

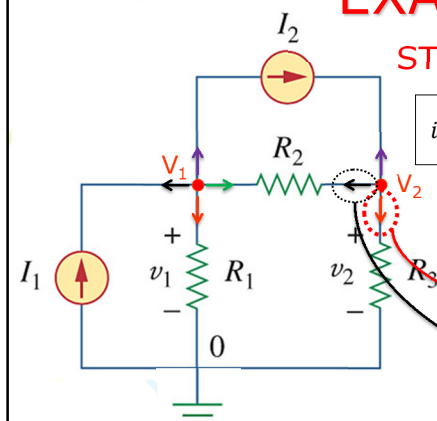
$$i = \frac{V_{High} - V_{Low}}{R_{Branch}}$$

- Number of critical nodes = number of equations
- Number of branches = number of terms.

9

3.2 Nodal Analysis

EXAMPLE 1



STEP 3: Apply KCL to each node

$$i = \frac{V_{High} - V_{Low}}{R_{Branch}}$$

$$i_{branch} = \frac{V_{n_from} - V_{n_to}}{R_{branch}}$$

Flows from?
Flows to?
Flows through?

V_2
Gnd
 R_3

V_2
 V_1
 R_2

$$i = \frac{V_2 - 0}{R_3}$$

$$i = \frac{V_2 - V_1}{R_2}$$

Node 2 (3 Branches):

$$\frac{V_2 - 0}{R_3} + \frac{V_2 - V_1}{R_2} + (\ominus I_2) = 0$$

Negative because outgoing branch current is opposite to the source's current

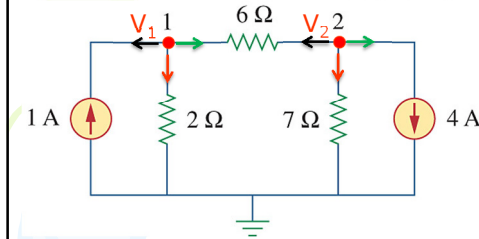
Node 1 (4 Branches):

$$\frac{V_1 - 0}{R_1} + \frac{V_1 - V_2}{R_2} + (I_2) + (\ominus I_1) = 0$$

10

3.2 Nodal Analysis

EXAMPLE 2



Node 1: $-1 + \frac{V_1 - 0}{2} + \frac{V_1 - V_2}{6} = 0$

x6

$$-6 + 3V_1 + V_1 - V_2 = 0$$

Write ito: $a_1V_1 + b_1V_2 = k_1$

$$4V_1 + (-1)V_2 = 6 \quad (1)$$

$$4V_1 + (-1)V_2 = 6 \quad (1)$$

$$(-7)V_1 + 13V_2 = -168 \quad (2)$$

$$a_1 = 4$$

$$b_1 = -1$$

$$a_2 = -7$$

$$b_2 = 13$$

Node 2: $\frac{V_2 - V_1}{6} + \frac{V_2 - 0}{7} + 4 = 0$

x42

$$7V_2 - 7V_1 + 6V_2 + 168 = 0$$

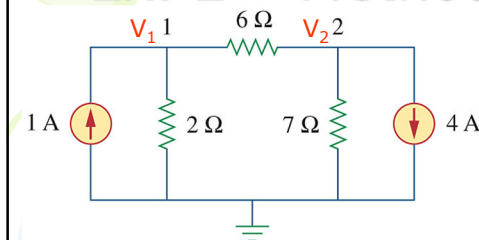
Write ito: $a_2V_1 + b_2V_2 = k_2$

$$(-7)V_1 + 13V_2 = -168 \quad (2)$$

11

3.2 Nodal Analysis

Ex. 2 – Method 1: Elimination



$$4V_1 + (-1)V_2 = 6 \quad (1)$$

$$(-7)V_1 + 13V_2 = -168 \quad (2)$$

From (1): $V_2 = 4V_1 - 6 \quad (3)$

(3) in (2): $(-7)V_1 + 13(4V_1 - 6) = -168$

$$45V_1 - 78 = -168$$

$$45V_1 = -90$$

$$\underline{V_1 = -2 \text{ V}}$$

From (3): $V_2 = 4(-2) - 6$

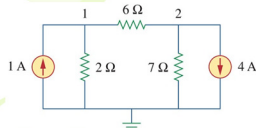
$$\underline{V_2 = -14 \text{ V}}$$

12

3.2 Nodal Analysis

Ex. 2 – Method 2: Cramer's Rule

See Appendix A



$$4V_1 + (-1)V_2 = 6 \quad (1)$$

$$(-7)V_1 + 13V_2 = -168 \quad (2)$$

$$\begin{bmatrix} 4 & -1 \\ -7 & 13 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 6 \\ -168 \end{bmatrix}$$

$$V_1 = \frac{\Delta_1}{\Delta} = \frac{-90}{45} = -2 \text{ V}$$

$$V_2 = \frac{\Delta_2}{\Delta} = \frac{-630}{45} = -14 \text{ V}$$

$$\Delta = \begin{vmatrix} 4 & -1 \\ -7 & 13 \end{vmatrix}$$

$$\Delta = (4 \times 13) - (-1 \times -7)$$

$$\Delta = 45$$

$$\Delta_1 = \begin{vmatrix} 6 & -1 \\ -168 & 13 \end{vmatrix}$$

$$\Delta_1 = (6 \times 13) - (-1 \times -168)$$

$$\Delta_1 = -90$$

$$\Delta_2 = \begin{vmatrix} 4 & 6 \\ -7 & -168 \end{vmatrix}$$

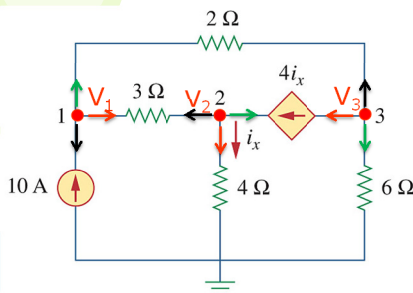
$$\Delta_2 = (4 \times -168) - (6 \times -7)$$

$$\Delta_2 = -630$$

13

PP 3-2

Determine the node voltages.



$$\begin{bmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 60 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Node 1: } -10 + \frac{V_1 - V_2}{3} + \frac{V_1 - V_3}{2} = 0$$

$$\times 6 \quad -60 + 2V_1 - 2V_2 + 3V_1 - 3V_3 = 0$$

$$5V_1 - 2V_2 - 3V_3 = 60 \quad (1)$$

$$\text{Node 2: } \frac{V_2 - V_1}{3} + \frac{V_2 - 0}{4} - 4i_x = 0$$

$$\times 12 \quad 4V_2 - 4V_1 + 3V_2 - 48 \frac{V_2}{4} = 0$$

$$-4V_1 - 5V_2 + 0V_3 = 0 \quad (2)$$

$$\text{Node 3: } 4i_x + \frac{V_3 - V_1}{2} + \frac{V_3 - 0}{6} = 0$$

$$\times 6 \quad 24 \frac{V_2}{4} + 3V_3 - 3V_1 + V_3 = 0$$

$$-3V_1 + 6V_2 + 4V_3 = 0 \quad (3)$$

14

PP 3-2 continued: Calculating Δ

$$\begin{bmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 60 \\ 0 \\ 0 \end{bmatrix}$$

$$V_1 = \frac{\Delta_1}{\Delta}$$

$$V_2 = \frac{\Delta_2}{\Delta}$$

$$V_3 = \frac{\Delta_3}{\Delta}$$

$$\Delta = \begin{vmatrix} +5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{vmatrix}$$

- + -
- + -
- + -

$$\begin{aligned} \Delta &= (5 \cdot -5 \cdot 4) + (-4 \cdot 6 \cdot -3) + (-3 \cdot -2 \cdot 0) \\ &\quad - (-3 \cdot -5 \cdot -3) - (0 \cdot 6 \cdot 5) - (4 \cdot -2 \cdot -4) \\ &= -100 + 72 + 0 - (-45) - 0 - 32 \\ &= -15 \end{aligned}$$

15

PP 3-2 continued: Calculating Δ_1

$$\begin{bmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 60 \\ 0 \\ 0 \end{bmatrix}$$

$$V_1 = \frac{\Delta_1}{\Delta} = \frac{-1200}{-15} = 80 \text{ V}$$

$$V_2 = \frac{\Delta_2}{\Delta}$$

$$V_3 = \frac{\Delta_3}{\Delta}$$

$$\Delta_1 = \begin{vmatrix} 60 & -2 & -3 \\ 0 & -5 & 0 \\ 0 & 6 & 4 \end{vmatrix}$$

- + -
- + -
- + -

$$\begin{aligned} \Delta_1 &= (60 \cdot -5 \cdot 4) + (0 \cdot 6 \cdot -3) + (0 \cdot -2 \cdot 0) \\ &\quad - (-3 \cdot -5 \cdot 0) - (0 \cdot 6 \cdot 60) - (4 \cdot -2 \cdot 0) \\ &= -1200 + 0 + 0 - 0 - 0 - 0 \\ &= -1200 \end{aligned}$$

16

PP 3-2 continued: Calculating Δ_2

$$\begin{bmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 60 \\ 0 \\ 0 \end{bmatrix}$$

$$V_1 = \frac{\Delta_1}{\Delta} = 80 \text{ V}$$

$$V_2 = \frac{\Delta_2}{\Delta} = \frac{960}{-15} = -64 \text{ V}$$

$$V_3 = \frac{\Delta_3}{\Delta}$$

$$\Delta_2 = \begin{vmatrix} +5 & 60 & -3 \\ -4 & 0 & 0 \\ -3 & 0 & +4 \end{vmatrix}$$

$$\begin{aligned} \Delta_2 &= (5 \cdot 0 \cdot 4) + (-4 \cdot 0 \cdot -3) + (-3 \cdot 60 \cdot 0) \\ &\quad - (-3 \cdot 0 \cdot -3) - (0 \cdot 0 \cdot 5) - (4 \cdot 60 \cdot -4) \\ &= 0 + 0 + 0 - 0 - 0 - (-960) \\ &= 960 \end{aligned}$$

17

PP 3-2 continued: Calculating Δ_3

$$\begin{bmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 60 \\ 0 \\ 0 \end{bmatrix}$$

$$V_1 = \frac{\Delta_1}{\Delta} = 80 \text{ V}$$

$$V_2 = \frac{\Delta_2}{\Delta} = -64 \text{ V}$$

$$V_3 = \frac{\Delta_3}{\Delta} = \frac{-2340}{-15} = 156 \text{ V}$$

$$\Delta_3 = \begin{vmatrix} +5 & -2 & 60 \\ -4 & -5 & 0 \\ -3 & +6 & 0 \end{vmatrix}$$

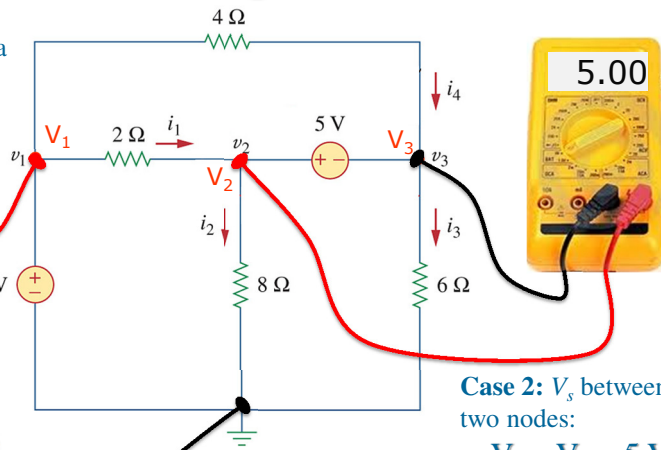
$$\begin{aligned} \Delta_3 &= 0 + (-4 \cdot 6 \cdot 60) + 0 \\ &\quad - (60 \cdot -5 \cdot -3) - 0 - 0 \\ &= -1440 - 900 \\ &= -2340 \end{aligned}$$

18

Ch. 3 L#2: 3.3 Nodal Analysis With Voltage sources

Case 1: V_s between a node and GND:

$$V_1 - 0 = 10 \text{ V}$$



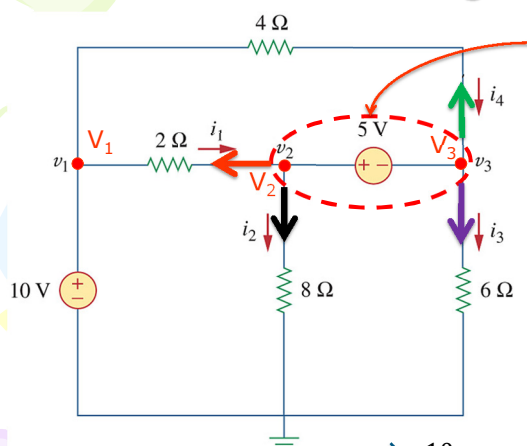
Case 2: V_s between two nodes:

$$V_2 - V_3 = 5 \text{ V}$$

SUPERNODE

19

3.3 Nodal Analysis With Voltage sources



SUPERNODE

Formed by enclosing a (dependent or independent) **voltage** source connected between two non-reference nodes along with any elements connected in **parallel** with that voltage source.

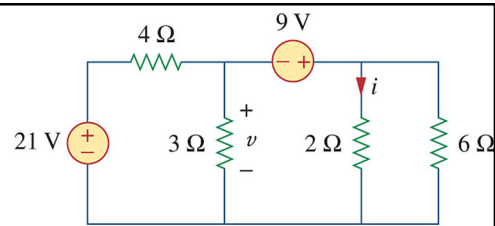
$$V_1 = 10 \quad (1)$$

$$V_2 - V_3 = 5 \quad (2)$$

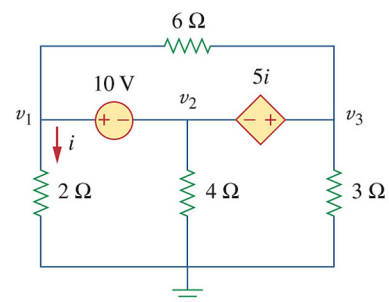
$$\text{KCL at SUPERNODE: } \frac{V_2 - V_1}{2} + \frac{V_2 - 0}{8} + \frac{V_3 - V_1}{4} + \frac{V_3 - 0}{6} = 0 \quad (3)$$

Solve for V_2 and V_3 from (2) and (3)

20

PP 3-3Find v and i in the circuit.

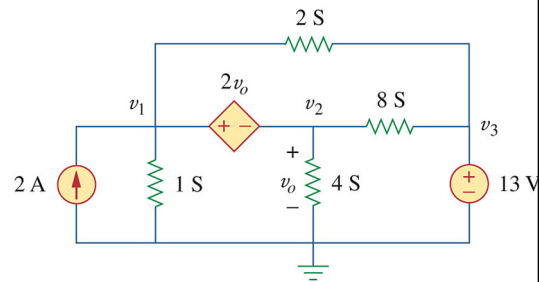
21

PP 3-4Find v_1 , v_2 , v_3 and i .

22

Problem 3-16

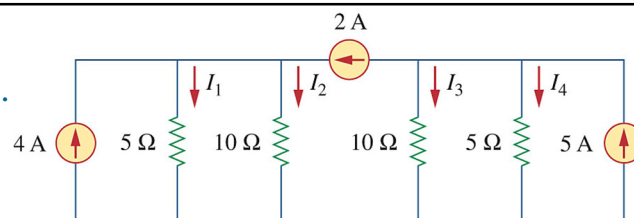
Find v_1 , v_2 and v_3 .



23

Problem 3-4

Find I_1 through I_4 .



24

Ch. 3 L#3: 3.4 Mesh Analysis

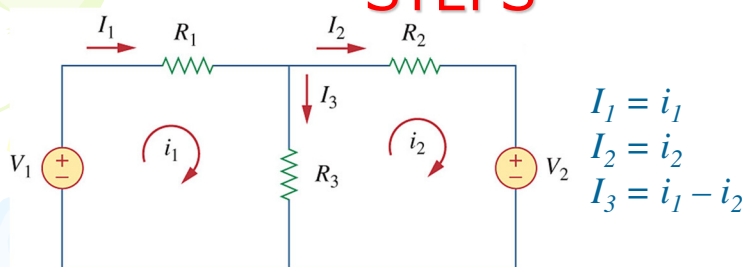
Why?

1. Mesh analysis provides another general procedure for analyzing circuits using mesh currents as the circuit variables.
2. Nodal analysis applies KCL to find unknown voltages in a given circuit, while mesh analysis applies KVL to find unknown currents.
3. A mesh is a loop which does not contain any other loops within it.

25

3.4 Mesh Analysis

STEPS



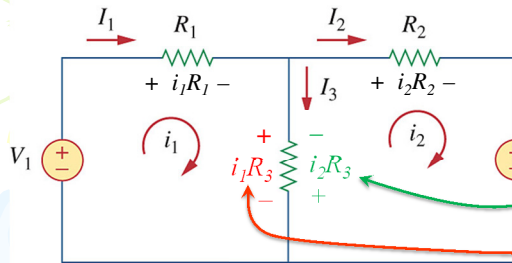
Steps to determine the mesh currents:

1. Assign mesh currents $i_1, i_2, \dots$, in to the n meshes – clock wise.
2. Apply KVL to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
3. Solve the resulting n simultaneous equations to get the mesh currents.

26

3.4 Mesh Analysis

STEPS: KVL in Loops



- R_3 has 2 currents flowing through it: i_1 and i_2
- i_2 causes a voltage drop from bottom to top: $i_2 R_3$
- i_1 causes a voltage drop from top to bottom: $i_1 R_3$

Loop 1: $-V_1 + R_1 i_1 + R_3 (i_1 - i_2) = 0$

Write ito: $a_1 i_1 + b_1 i_2 = k_1$

$(R_1 + R_3) i_1 + (-R_3) i_2 = V_1$

I_{net} through R_3
wrt loop 1

$R(i_{loop} - i_{opposing_loop})$

Loop 2: $V_2 + R_3 (i_2 - i_1) + i_2 R_2 = 0$

Write ito: $a_2 i_1 + b_2 i_2 = k_2$

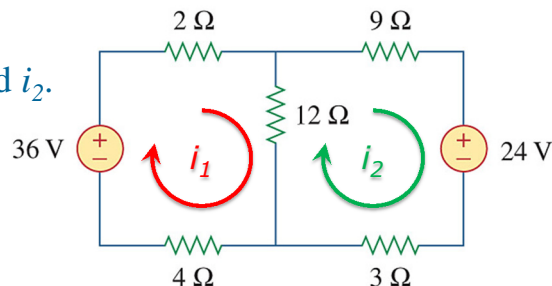
$(-R_3) i_1 + (R_2 + R_3) i_2 = -V_2$

I_{net} through R_3
wrt loop 2

27

PP 3-5

Determine i_1 and i_2 .



Loop 1: $-36 + 2i_1 + 12(i_1 - i_2) + 4i_1 = 0$

$(2+12+4)i_1 + (-12)i_2 = 36$

$(18)i_1 + (-12)i_2 = 36$ (1)

Loop 2: $24 + 3i_2 + 12(i_2 - i_1) + 9i_2 = 0$

$(-12)i_1 + (3+12+9)i_2 = -24$

$(-12)i_1 + (24)i_2 = -24$ (2)

Matrix: $\begin{bmatrix} 18 & -12 \\ -12 & 24 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 36 \\ -24 \end{bmatrix}$

$\Delta = \begin{vmatrix} 18 & -12 \\ -12 & 24 \end{vmatrix} = 288$

$\Delta_1 = \begin{vmatrix} 36 & -12 \\ -24 & 24 \end{vmatrix} = 576$

$\Delta_2 = \begin{vmatrix} 18 & 36 \\ -12 & -24 \end{vmatrix} = 0$

$i_1 = \frac{\Delta_1}{\Delta} = \frac{576}{288} = 2 \text{ A}$

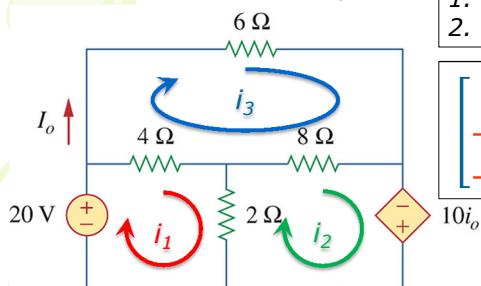
$i_2 = \frac{\Delta_2}{\Delta} = \frac{0}{288} = 0 \text{ A}$

28

PP 3-6 Determine I_o .

Note:

1. $I_o = i_3$
2. CCVS: $10i_o = 10i_3$



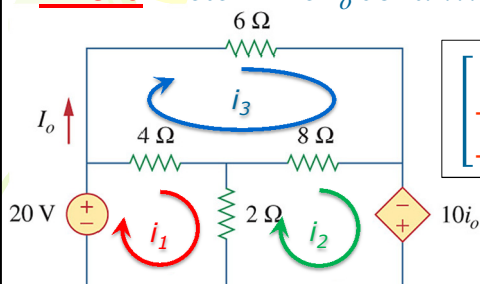
$$\begin{bmatrix} 6 & -2 & -4 \\ -2 & 10 & -18 \\ -4 & -8 & 18 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 0 \\ 0 \end{bmatrix}$$

Loop 1: $-20 + 4(i_1 - i_3) + 2(i_1 - i_2) = 0$
 $(4+2)i_1 + (-2)i_2 + (-4)i_3 = 20$
 $6i_1 + (-2)i_2 + (-4)i_3 = 20$ (1)

Loop 2: $-10i_3 + 2(i_2 - i_1) + 8(i_2 - i_3) = 0$
 $(-2)i_1 + (10)i_2 + (-18)i_3 = 0$ (2)

Loop 3: $6i_3 + 8(i_3 - i_2) + 4(i_3 - i_1) = 0$
 $(-4)i_1 + (-8)i_2 + (18)i_3 = 0$ (3)

29

PP 3-6 Determine I_o cont. ...

$$\begin{bmatrix} 6 & -2 & -4 \\ -2 & 10 & -18 \\ -4 & -8 & 18 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 0 \\ 0 \end{bmatrix}$$

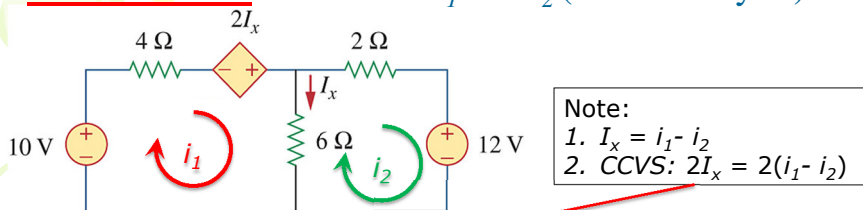
$$\Delta = \begin{vmatrix} 6 & -2 & -4 \\ -2 & 10 & -18 \\ -4 & -8 & 18 \end{vmatrix}$$

$$\begin{aligned} \Delta &= (6 \cdot 10 \cdot 18) + (-2 \cdot -8 \cdot -4) + (-4 \cdot -2 \cdot -18) \\ &\quad - (-4 \cdot 10 \cdot -4) - (-18 \cdot -8 \cdot 6) - (18 \cdot -2 \cdot -2) \\ &= 1080 - 64 - 144 - 160 - 864 - 72 \\ &= -224 \end{aligned}$$

$$\Delta_3 = 0 + 320 + 0 - (-800) - 0 - 0 = 1120$$

$$I_o = i_3 = \frac{\Delta_3}{\Delta} = \frac{1120}{-224} = -5 \text{ A}$$

$$\Delta_3 = \begin{vmatrix} 6 & -2 & 20 \\ -2 & 10 & 0 \\ -4 & -8 & 0 \end{vmatrix}$$

Problem 3-39 Determine i_1 and i_2 (Mesh Analysis)

Note:

1. $I_x = i_1 - i_2$
2. CCVS: $2I_x = 2(i_1 - i_2)$

Loop 1: $-10 + 4i_1 - 2(i_1 - i_2) + 6(i_1 - i_2) = 0$

$$8i_1 + (-4)i_2 = 10 \quad (1)$$

Loop 2: $12 + 6(i_2 - i_1) + 2i_2 = 0$

$$(-6)i_1 + 8i_2 = -12 \quad (2)$$

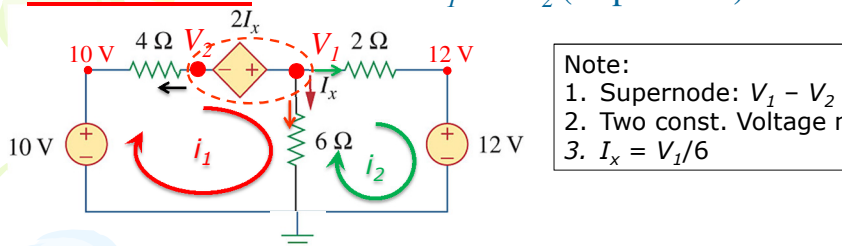
$$(2) + 2 \times (1): (-6+16)i_1 + (8-8)i_2 = (-12+20)$$

$$10i_1 + 0i_2 = 8$$

$$\therefore i_1 = 0.8 \text{ A} \quad (3)$$

$$(3) \text{ in } (2): (-6)(0.8) + 8i_2 = -12 \quad \therefore i_2 = -0.9 \text{ A}$$

31

Problem 3-39 Determine i_1 and i_2 (Supernode)

Note:

1. Supernode: $V_1 - V_2 = 2I_x$
2. Two const. Voltage nodes
3. $I_x = V_1/6$

KCL @ SN: $\frac{V_2 - 10}{4} + \frac{V_1}{6} + \frac{V_1 - 12}{2} = 0$

$$\times 12 \quad 3V_2 - 30 + 2V_1 + 6V_1 - 72 = 0$$

$$8V_1 + 3V_2 = 102 \quad (1)$$

Supernode: $V_1 - V_2 = 2I_x = 2 \cdot \frac{V_1}{6} = \frac{V_1}{3}$

$$\times 3 \quad 3V_1 - 3V_2 = V_1$$

$$2V_1 - 3V_2 = 0 \quad (2)$$

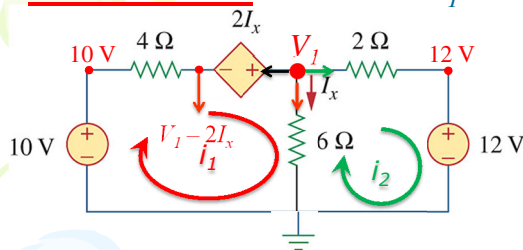
$$i_1 = \frac{10 - V_2}{4} = \frac{3.2}{4} = 0.8 \text{ A}$$

$$i_2 = \frac{V_1 - 12}{2} = \frac{10.2 - 12}{2} = -0.9 \text{ A}$$

$$(1) + (2): (8+2)V_1 + (3-3)V_2 = 102 \quad \therefore V_1 = 10.2 \text{ V}$$

$$\therefore V_2 = 6.8 \text{ V}$$

32

Problem 3-39 Determine i_1 and i_2 (Node Voltage)

Note:

1. Only 1 critical node
2. Const. Voltage nodes
3. $I_x = V_1/6$
4. V over $4\ \Omega$

Node 1: $\frac{(V_1 - 2I_x) - 10}{4} + \frac{V_1}{6} + \frac{V_1 - 12}{2} = 0$

x12 $3V_1 - 6I_x - 30 + 2V_1 + 6V_1 - 72 = 0$

$3V_1 - 6\frac{V_1}{6} - 30 + 2V_1 + 6V_1 - 72 = 0$

$10V_1 = 102$

$V_1 = 10.2$

$i_1 = - \left[\frac{(V_1 - 2V_1/6) - 10}{4} \right]$

$= - \left[\frac{(10.2 - 20.4/6) - 10}{4} \right]$

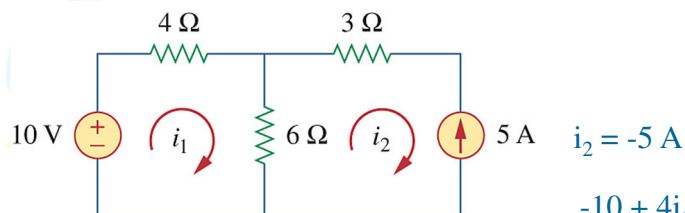
$= 0.8\text{ A}$

$i_2 = \frac{V_1 - 12}{2} = \frac{10.2 - 12}{2} = -0.9\text{ A}$

33

Ch. 3 L#4: 3.5 Mesh Analysis With Current Sources

CASE #1: A current source exists only in one mesh:
That loop current is then known.



$-10 + 4i_1 + 6(i_1 - i_2) = 0$

$-10 + 10i_1 - 6(-5) = 0$

$10i_1 = -20$

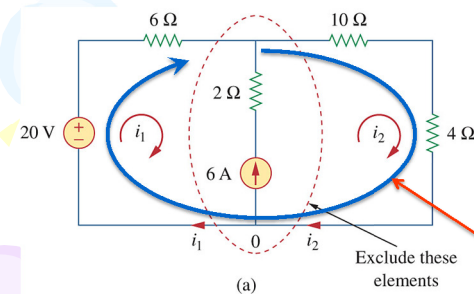
$i_1 = -2\text{ A}$

34

3.5 Mesh Analysis With Current Sources

CASE #2: A current source exists between two meshes: SUPERMESH

A **SUPERMESH** results when two meshes have a current source (dependent or independent), with or without other elements in series, in common.



Common source:

Note: i_2 flows in the same direction as the source.

$$\begin{aligned} i_2 - i_1 &= 6 \\ i_1 &= i_2 - 6 \end{aligned} \quad (1)$$

SUPERMESH:

$$-20 + 6i_1 + 10i_2 + 4i_2 = 0 \quad (2)$$

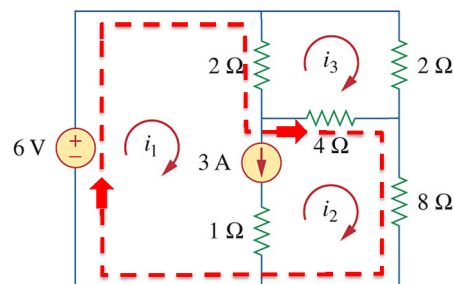
(1) in (2) $-20 + 6(i_2 - 6) + 10i_2 + 4i_2 = 0$

$$i_2 = 2.8 \text{ A}$$

$$\therefore i_1 = -3.2 \text{ A}$$

35

PP 3-7 Determine i_1 , i_2 and i_3 .



Common source:

$$i_1 - i_2 = 3 \quad (1)$$

Supermesh:

$$-6 + 2(i_1 - i_3) + 4(i_2 - i_3) + 8i_2 = 0$$

$$2i_1 + 12i_2 - 6i_3 = 6 \quad (2)$$

Loop 3:

$$2(i_3 - i_1) + 2i_3 + 4(i_3 - i_2) = 0$$

$$-2i_1 - 4i_2 + 8i_3 = 0 \quad (3)$$

36

Problem 3-46

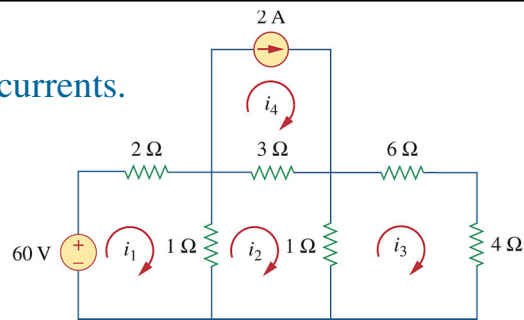
Solve for the mesh currents.

$$i_4 = 2 \text{ A}$$

Loop 1:

Loop 2:

Loop 3:



37

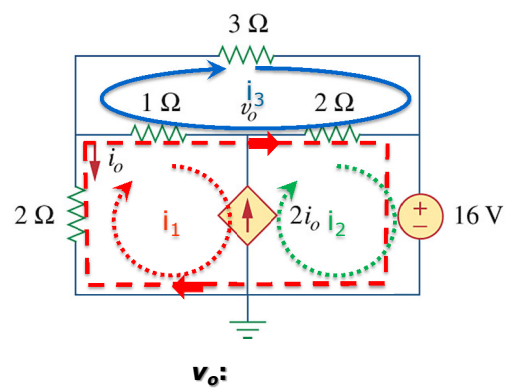
Problem 3-46

Find v_o and i_o

Common source:

Supermesh:

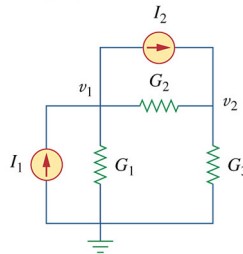
Loop 3:



38

Ch. 3 L#5: 3.6 Nodal and Mesh Analysis by inspection

Special case #1: All sources = independent **current** sources



$$\begin{matrix} \textcircled{1} & \textcircled{2} \\ \textcircled{1} & \textcircled{2} \end{matrix} \begin{bmatrix} G_1 + G_2 & -G_2 \\ -G_2 & G_2 + G_3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} I_1 - I_2 \\ I_2 \end{bmatrix}$$

Conductance Matrix

$\begin{bmatrix} \blacksquare & \\ & \blacksquare \end{bmatrix}$ - Diagonal elements ($G_{11}, G_{22}, \dots$) = sum of all G 's connected to that node.

$\begin{bmatrix} & \blacksquare \\ \blacksquare & \end{bmatrix}$ - Off-diagonal elements = $-G$ connected between the two nodes.

Sum of the currents entering the node:

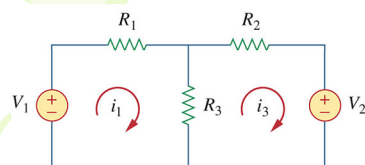
In = +

Out = -

39

3.6 Nodal and Mesh Analysis by inspection

Special case #2: All sources = independent **voltage** sources



$$\begin{matrix} \textcircled{1} & \textcircled{2} \\ \textcircled{1} & \textcircled{2} \end{matrix} \begin{bmatrix} R_1 + R_3 & -R_3 \\ -R_3 & R_2 + R_3 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ -V_2 \end{bmatrix}$$

Resistance Matrix

$\begin{bmatrix} \blacksquare & \\ & \blacksquare \end{bmatrix}$ - Diagonal elements ($R_{11}, R_{22}, \dots$) = sum of all R 's in the loop.

$\begin{bmatrix} & \blacksquare \\ \blacksquare & \end{bmatrix}$ - Off-diagonal elements = $-R$ common to two loop between the two nodes

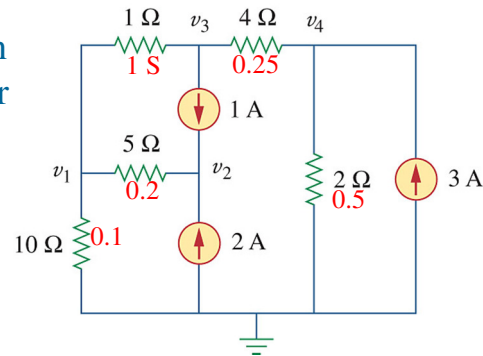
Sum of all the voltage increases in the loop:

In @ - = +

In @ + = -

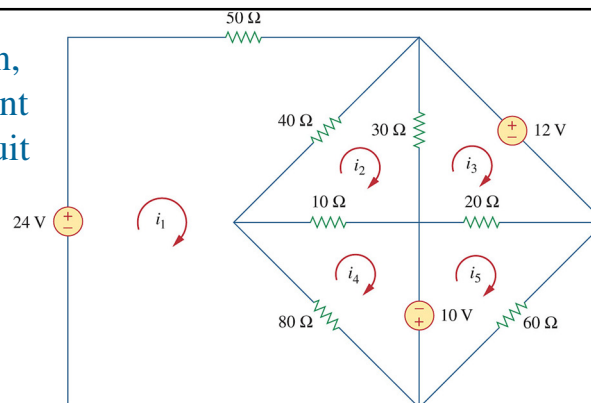
Remember: V is taken to the other side in the KVL equation.

PP 3-8 By inspection, obtain the node voltage equations for the circuit in matrix form.



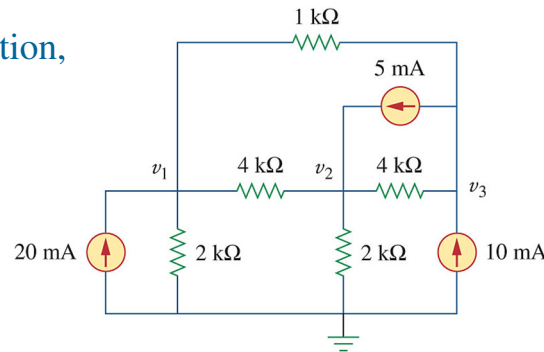
41

PP 3-9 By inspection, obtain the mesh-current equations for the circuit in matrix form.



42

Problem 3-69 By inspection, obtain the node voltage equations for the circuit



43

3.7 Nodal versus Mesh Analysis (1)

To select the method that results in the smaller number of equations. For example:

1. Choose nodal analysis for circuit with fewer nodes than meshes.
 - *Choose mesh analysis for circuit with fewer meshes than nodes.
 - *Networks that contain many series connected elements, voltage sources, or supermeshes are more suitable for mesh analysis.
 - *Networks with parallel-connected elements, current sources, or supernodes are more suitable for nodal analysis.
2. If node voltages are required, it may be expedient to apply nodal analysis. If branch or mesh currents are required, it may be better to use mesh analysis.

44

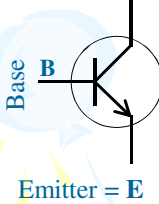


Ch. 3 L#6:

3.9 Transistor Circuits

BJT = Bipolar Junction Transistor



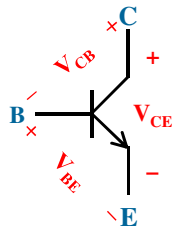


Collector = C

Base = B

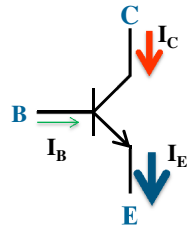
Emitter = E

npn transistor



KVL: $V_{CE} = V_{BE} + V_{CB}$
 $V_{BE} = 0.7 \text{ V}$

Voltages



KCL: $I_E = I_C + I_B$

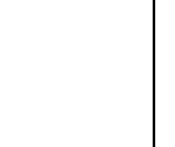
Currents

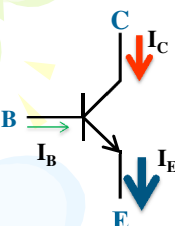
45



3.9 Transistor Circuits

Currents





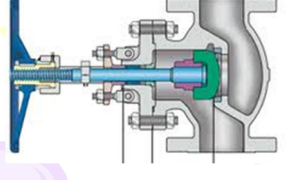
KCL: $I_E = I_C + I_B$
 $I_C = \beta I_B$
 $\therefore I_E = (1 + \beta) I_B$

$I_C = \alpha I_E$

$$\beta = \frac{\alpha}{1 - \alpha}$$

β = “Common-emitter current gain”
 β ranges between 50 $\rightarrow$ 100

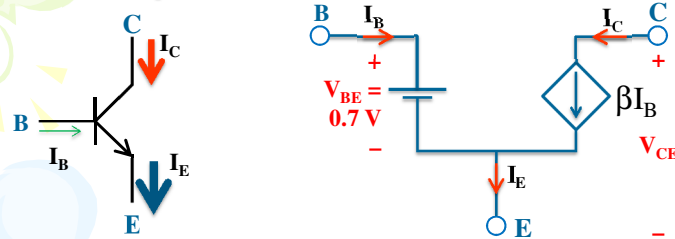
α = “Common-base current gain”
 α ranges between 0.98 $\rightarrow$ 0.999



46

3.9 Transistor Circuits

Equivalent network model



A transistor is equivalent to a current controlled current source.

How does one analyse a circuit with a transistor in it?

Method #1: Mesh Analysis

Method #2: Node Analysis
Only applicable if the equivalent network model is used.

47

PP 3-12 Determine v_o and V_{CE} for the transistor circuit in which $V_{BE} = 0.7$ V and $\beta = 100$. **METHOD #1**

KVL for Input Loop:

$$-5 + 10000 \cdot I_B + 0.7 + 200 \cdot I_E = 0 \quad (1)$$

KVL for Output Loop:

$$-200 \cdot I_E - V_{CE} - 500 \cdot I_C + 12 = 0 \quad (2)$$

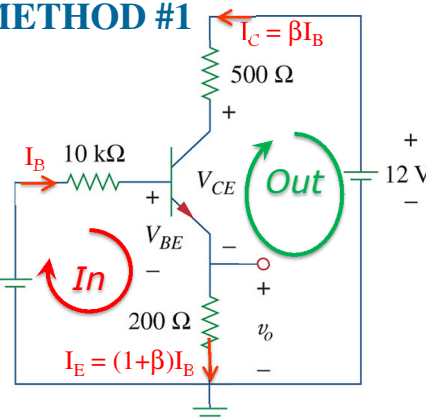
Note:

$$I_C = \beta I_B$$

$$v_o = 200 \times I_E = 200 \times (1 + \beta) I_B$$

The Method:

1. Write loop equations in terms of currents I_B , I_E and I_C .
2. Use $I_E = (1 + \beta) I_B$ and $I_C = \beta I_B$ to write all in terms of I_B .
3. Solve for I_B from Input loop.
4. Solve for V_{CE} from Output loop.



48

PP 3-12 continued...**From (1) solve for I_B :**

$$-5 + 10000 \cdot I_B + 0.7 + 200(1+100)I_B = 0$$

$$30200 \cdot I_B = 4.3$$

$$I_B = 0.1424 \text{ mA}$$

Hence:

$$v_o = 200 \times (1+100)I_B$$

$$= 200 \times 101 \cdot 0.142 \times 10^{-3}$$

$$= 2.87 \text{ V}$$

From (2) solve for:

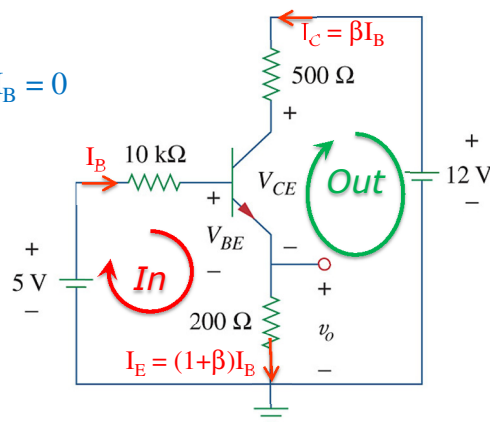
$$-v_o - V_{CE} - 500 \cdot I_C + 12 = 0$$

$$-[200 \times (1+\beta)I_B] - V_{CE} - 500 \cdot \beta I_B + 12 = 0$$

$$V_{CE} = 12 - 20200 \cdot I_B - 50000 \cdot I_B$$

$$= 12 - 70200 \cdot I_B$$

$$= 2 \text{ V}$$

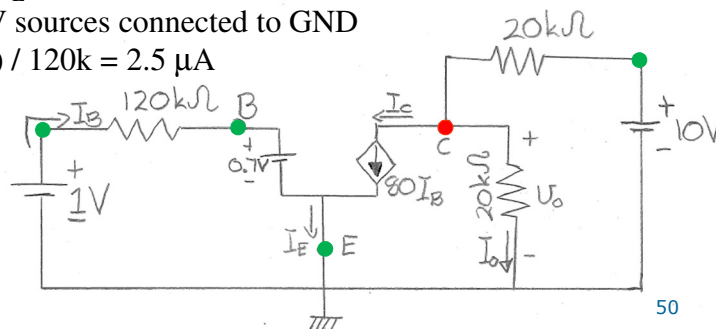
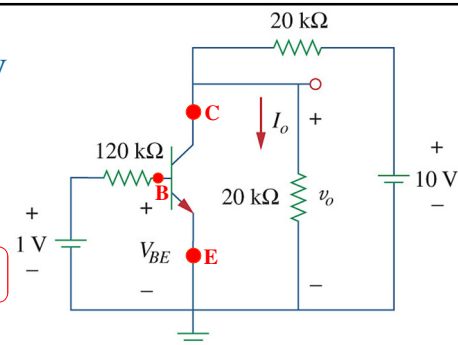


49

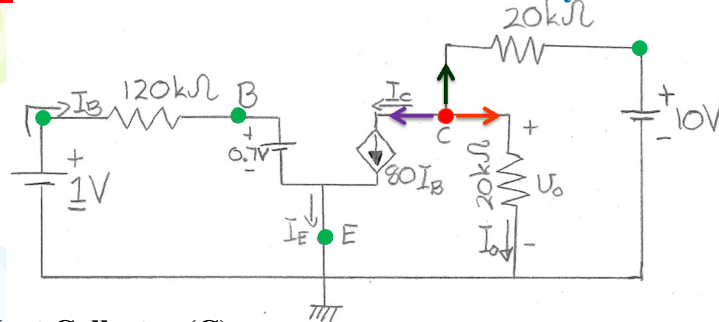
PP 3-13 Determine v_o and I_o for the transistor circuit in which $V_{BE} = 0.7 \text{ V}$ and $\beta = 80$.

METHOD #2 – Node Analysis**Note:**

- E is connected to GND @ 0 V
- $V_{CE} = V_C - V_E = V_C - 0 = V_C = v_o$
- $I_o = v_o / 20\text{k}$
- $V_{BE} = V_B - V_E = 0.7 - 0 = 0.7 \text{ V}$
- 1 V and 10 V sources connected to GND
- $I_B = (1 - 0.7) / 120\text{k} = 2.5 \mu\text{A}$



50

PP 3-13 continued... **METHOD #2 – Node Analysis****KCL at Collector (C):**

$$80I_B + \frac{v_o}{20000} + \frac{v_o - 10}{20000} = 0$$

$$20000 \times 80 \left(\frac{1 - 0.7}{120000} \right) + 2v_o - 10 = 0$$

$$2v_o = 10 - 4$$

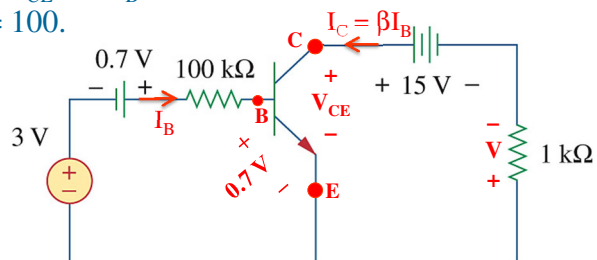
$$v_o = 3 \text{ V}$$

$$I_o = \frac{v_o}{20000} = \frac{3}{20000} = 150 \mu\text{A}$$

51

Problem 3-89 Determine V_{CE} and I_B for the transistor circuit in which $V_{BE} = 0.7 \text{ V}$ and $\beta = 100$.**Note:**

- E @ 0 V
- $V_{BE} = V_B = 0.7 \text{ V}$



52